4.1 Studying Cells

A cell is the smallest unit of a living thing, and every organism, whether it consists of a single cell, like a bacterium, or of trillions of cells, like a human being, is constructed from these fundamental units, which carry out all of the chemical reactions that are necessary for life.

Because most cells are far too small to be seen with the unaided eye, the study of cells depended on the development of the microscope; in 1665 Robert Hooke examined thin slices of cork and described the tiny compartments he observed as cells, a term that biologists continue to use today.

Light microscopes, which pass visible light through a specimen and bend it through a system of lenses, can magnify objects up to approximately 1,000 times, whereas electron microscopes, which use a beam of electrons rather than light, achieve magnifications of up to 1,000,000 times and therefore reveal structures inside the cell that cannot otherwise be resolved.

The cell theory, developed during the nineteenth century by Matthias Schleiden, Theodor Schwann and Rudolf Virchow, states that all living organisms are composed of one or more cells, that the cell is the basic unit of life, and that new cells arise only from the division of pre-existing cells.

4.2 Prokaryotic Cells

Prokaryotes, which include bacteria and archaea, are single-celled organisms that lack a membrane-bound nucleus and other membrane-bound organelles; instead, their genetic material is located in a central region of the cell called the nucleoid.

All prokaryotic cells share four common components: a plasma membrane that separates the interior of the cell from its environment, cytoplasm in which the cellular reactions occur, DNA that carries the genetic instructions, and ribosomes that synthesise proteins.

Most prokaryotes range from 0.1 to 5.0 micrometres in diameter, and their small size is significant because it allows ions and organic molecules that enter the cell to diffuse rapidly to all of its regions, while waste products produced within the cell can likewise diffuse out quickly.

As a cell increases in size, its volume grows faster than its surface area; for a cube with side length L, the ratio is expressed as SA:V = 6/L, which means that a larger cell has relatively less membrane through which to exchange materials with its surroundings.